

EduVision – Education Analytics Methodology

Data preparation, KPI engineering, modeling, and validation methodology

1. Analytical Methodology

EduVision follows a structured data analytics workflow: source collection → integration → standardization → validation → KPI derivation → visualization → testing. The objective is to transform heterogeneous higher-education source data into a consistent dataset suitable for interactive analysis.

2. Data Collection

- Collect ranking and higher-education datasets from the project-approved source files.
- Inspect schemas, column names, data types, missing values, and key identifiers.
- Preserve raw inputs separately from processed outputs.

3. Data Integration

- Align university naming and other matching fields before integration.
- Use university and year context where available to reduce incorrect matches.
- Combine complementary ranking, research, student, faculty, and country-level attributes.
- Retain source-level fields needed for traceability and validation.

4. Data Cleaning and Standardization

- Standardize text fields such as university and country names.
- Handle missing, inconsistent, and duplicate records according to project rules.
- Convert numeric fields to appropriate numeric types.
- Check percentage, ratio, rank, and score ranges.

5. KPI Engineering

- Generate research and productivity indicators required by the dashboard.
- Calculate country-level aggregates such as average rank and average performance scores.
- Create supporting counts and ratios for student and faculty analysis.
- Keep KPI calculations reproducible through documented Python transformation logic or transparent Power BI calculations.

6. Dashboard Modeling

- Load the final analytical dataset into Power BI.
- Create measures, calculated fields, filters, and visuals required by the four dashboard areas.
- Use consistent dimensions such as university, country, and year.

7. Validation

- Compare KPI results against source and intermediate calculations.

- Check rank ordering and country aggregation logic.
- Test filters and navigation across dashboard pages.
- Inspect extreme values and unexpected blanks.
- Perform final visual QA before delivery.

8. Limitations

- Ranking methodologies differ between source organizations; source scores should not automatically be treated as directly equivalent.
- Coverage can vary by university, country, year, and indicator.
- Country averages depend on which universities are represented in the dataset.
- Derived metrics are analytical indicators and should be interpreted within the documented methodology.

9. Recommended Reproduction Order

1) Place approved raw datasets in **data/raw** → 2) run collection/integration logic → 3) run cleaning/validation → 4) generate KPI fields → 5) export the final dataset → 6) refresh the Power BI dashboard → 7) run QA checks → 8) commit the final project to GitHub.